PYTHON WHILE LOOP — COMPLETE NOTES

Beginner-friendly notes with important concepts and 10 practical examples.

1. What is a while loop?

A while loop repeatedly executes a block of code as long as its condition is True. It is useful when the number of repetitions is not known in advance.

2. Basic syntax

```
while condition:
    # statements
```

Example:

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

3. Important points

• The condition is checked before every iteration.
• If the condition is False initially, the loop may execute zero times.
• Update the loop-control variable to avoid an infinite loop.
• break exits the loop immediately.
• continue skips the current iteration.
• pass is a placeholder and performs no action.

4. Infinite loop

An infinite loop continues forever because its condition never becomes False. Use it carefully.

```
i = 1
while i <= 5:
    print(i)
    i += 1

# Without i += 1, the loop could continue forever.
```

5. while loop with else

Python allows an else block with while. It runs when the loop finishes normally, but not when the loop is terminated by break.

```
i = 1
while i <= 3:
    print(i)
    i += 1
else:
    print("Loop completed")
```

6. break and continue

```
i = 1
while i <= 5:
    if i == 4:
        break
    print(i)
    i += 1
```

```
i = 0
while i < 5:
    i += 1
    if i == 3:
        continue
    print(i)
```

WHILE LOOP — 10 EXAMPLES

Example 1: Print numbers 1 to 10

```
i = 1
while i <= 10:
    print(i)
    i += 1
```

Example 2: Print even numbers from 1 to 20

```
i = 2
while i <= 20:
    print(i)
    i += 2
```

Example 3: Print odd numbers from 1 to 20

```
i = 1
while i <= 20:
    print(i)
    i += 2
```

Example 4: Sum of numbers 1 to 10

```
i = 1
total = 0

while i <= 10:
    total += i
    i += 1

print("Sum =", total)
```

Example 5: Multiplication table

```
n = 5
i = 1

while i <= 10:
    print(n, "x", i, "=", n * i)
    i += 1
```

Example 6: Count digits in a number

```
n = 12345
count = 0

while n > 0:
    n //= 10
    count += 1

print("Digits =", count)
```

Example 7: Reverse a number

```
n = 12345
reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n //= 10

print("Reverse =", reverse)
```

Example 8: Check palindrome number

```
n = 121
original = n
reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n //= 10

if original == reverse:
    print("Palindrome")
else:
    print("Not palindrome")
```

Example 9: Factorial of a number

```
n = 5
fact = 1
i = 1

while i <= n:
    fact *= i
    i += 1

print("Factorial =", fact)
```

Example 10: Check prime number

```
n = 17
i = 1
count = 0

while i <= n:
    if n % i == 0:
        count += 1
    i += 1

if count == 2:
    print("Prime number")
else:
    print("Not a prime number")
```